

PSEUDO CORD WRITING

1.

```
#define DEBUG 0

IF DEBUG
    PRINT intermediate_value
END IF
```

Error:

IF DEBUG is incorrect because #define is a preprocessor directive. In C it should be:

```
#if DEBUG
printf("%d", intermediate_value);
#endif
```

Correct Logic:

Print intermediate_value only when DEBUG is 1.

2.

Fill Constant

```
#define PI 3.14
```

Fill Formula

```
area = PI * radius * radius
```

3.

```
#define RATE 1.5

DECLARE x AS int = 10
DECLARE result AS float

result = x * RATE

PRINT result
```

Output

15.0

Why is result float?

Because RATE is **1.5**, a floating-point constant. Multiplying an integer by a float gives a float result.

4.

```
#define MAX 1000

DECLARE temp AS int

INPUT temp

IF temp > MAX
    PRINT "Valid"
ELSE
    PRINT "Invalid"
```

Logical Error

The condition is wrong.

Correct Condition

```
IF temp <= MAX
    PRINT "Valid"
ELSE
    PRINT "Invalid"
END IF
```

5.

Arrange in correct order

1. DEFINE conversion rate
2. INPUT amount
3. CALCULATE converted value
4. PRINT result

6.

Missing control structure

```
#define MIN 10
#define MAX 50

DECLARE value AS int

INPUT value

IF (value >= MIN AND value <= MAX)
    PRINT "Within range"
ELSE
    PRINT "Out of range"
END IF
```

7.

Pseudocode

```
#define DISCOUNT_ENABLED 1
#define TAX_ENABLED 1

#define DISCOUNT 10
#define TAX 5

DECLARE price AS float
DECLARE quantity AS int
DECLARE total AS float

INPUT price
INPUT quantity

total = price * quantity

#if DISCOUNT_ENABLED
total = total - (total * DISCOUNT / 100)
#endif

#if TAX_ENABLED
total = total + (total * TAX / 100)
#endif

PRINT total
```

8.

Given

```
#define TAX 5%

DECLARE price AS int
DECLARE total AS int

total = price + (price * TAX)
```

Errors

- #define TAX 5% is invalid.
- price should be float.
- total should be float.
- Tax must be divided by 100.

Correct Version

```
#define TAX 5

DECLARE price AS float
DECLARE total AS float

INPUT price
```

```
total = price + (price * TAX / 100)

PRINT total
```

9.

Pseudocode

```
#define CELSIUS_TO_FAHRENHEIT 1
#define FAHRENHEIT_TO_CELSIUS 0

DECLARE temp AS float
DECLARE result AS float

INPUT temp

#if CELSIUS_TO_FAHRENHEIT
result = (temp * 9 / 5) + 32
#endif

#if FAHRENHEIT_TO_CELSIUS
result = (temp - 32) * 5 / 9
#endif

PRINT result
```

10.

Extend pseudocode

```
#define SIZE 5

DECLARE arr[SIZE]
DECLARE i AS int
DECLARE sum AS int = 0
DECLARE average AS float

FOR i = 0 TO SIZE-1
    INPUT arr[i]
    sum = sum + arr[i]
END FOR

average = sum / SIZE

PRINT sum
PRINT average
```